Early Autism Detection Using Repetitive Movement

Abstract:

The early diagnosis of autism has a critical role in the child's developmental growth since it leads to timely interventions that significantly improve the long-term outcomes. Nevertheless, conventional diagnostics is extremely time-consuming, subjective, and is heavily dependent on the expertise of a trained professional. This paper proposes a deep learning approach for early autism detection using short video clips of the children. In this study, the attention is given to identifying the repetitive behaviors of hand flapping, body rocking, and spinning exhibited among children aged greater than four years, which have been identified as early signs of ASD.

Drawing inspiration from the paper "Advanced Gesture Recognition for Autism Spectrum Disorder Detection," this research proposes a two-stage approach that integrates object detection and motion pattern analysis. First, the YOLOv7 model is used to detect the child in every frame of the video input, so that all unnecessary background distractions are excluded in the video processing step. The refined video will, in the next step, go through a VideoMAE (Masked Autoencoder) model that rebuilds missing parts in videos for better capture of motion continuity and dynamics related to repetitive gestures. The model yielded an accuracy of more than 94% for repetitive movement classification, which proved that this can be a potentially effective approach toward early identification. Its main advantage is that this method is non-subjective and consistent, avoiding human biases and expert dependencies. Besides, since it is purely a software solution, it could easily be installed on portable or even mobile devices for accessible, low-cost, early screening of autism in resource-constrained areas. Our long-term objective is to make early autism screening faster, more objective, and accessible. Facilitating timely detection and intervention, we hope this technology will help children get the much-needed support and care during the most critical years of their lives

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